习题　5-3

1.计算下列定积分.

(1) $\int_{-2}^{1}\frac{dx}{(11+5x)^3}$;　　(2) $\int_{0}^{\sqrt{2}}\sqrt{2-x^2}dx$;

(3) $\int_{\frac{\sqrt{2}}{2}}^{1}\frac{\sqrt{1-x^2}}{x^2}dx$;　　(4) $\int_{1}^{\sqrt{3}}\frac{1}{x^2\sqrt{1+x^2}}dx$;

(5) $\int_{-1}^{1}\frac{xdx}{\sqrt{5-4x}}$;　　(6) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos x\cos 2xdx$;

(7) $\int_{0}^{\frac{\pi}{2}}\frac{dx}{3+2\cos x}$;　　(8) $\int_{0}^{\frac{\pi}{2}}\frac{\cos x}{6-5\sin x+\sin^2 x}dx$;

(9) $\int_{0}^{\pi}(1-\sin^3 x)dx$;　　(10) $\int_{0}^{\frac{\pi}{4}}\frac{1-\cos^4 x}{2}dx$;

(11) $\int_{1}^{e^2}\frac{dx}{x\sqrt{1+\ln x}}$;　　(12) $\int_{1}^{\ln 2}\sqrt{e^x-1}dx$.

解(1) $\int_{-2}^{1}\frac{dx}{(11+5x)^3}=\frac{1}{5}\int_{-2}^{1}\frac{d(11+5x)}{(11+5x)^3}=-\frac{1}{10}\left[\frac{1}{(11+5x)^2}\right]_{-2}^{1}=-\frac{1}{10}\left[\frac{1}{256}-1\right]=\frac{51}{512}$;

(2) $\int_{0}^{\sqrt{2}}\sqrt{2-x^2}\mathrm{d}x\overset{x=\sqrt{2}\sin t}{=\!=\!=}\int_{0}^{\frac{\pi}{2}}2\cos^2 t\mathrm{d}t=2\cdot\frac{\pi}{4}=\frac{\pi}{2}$.

(3) $\int_{\frac{1}{\sqrt{2}}}^{1}\frac{\sqrt{1-x^2}}{x^2}\mathrm{d}x\overset{x=\sin t}{=\!=\!=}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{\cos^2 t}{\sin^2 t}\mathrm{d}t=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\left(\csc^2 x-1\right)\mathrm{d}t=\left[-\cot t-t\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}=1-\frac{\pi}{4}$.

(4) $\int_{1}^{\sqrt{3}}\frac{1}{x^2\sqrt{1+x^2}}dx\overset{x=\frac{1}{t}}{=\!=\!=}\int_{1}^{\frac{\sqrt{3}}{3}}\frac{1}{\left(\frac{1}{t}\right)^2\sqrt{1+\left(\frac{1}{t}\right)^2}}\left(-\frac{1}{t^2}\right)dt=-\int_{1}^{\frac{\sqrt{3}}{3}}\frac{t}{\sqrt{1+t^2}}dt$

$=-\frac{1}{2}\int_{1}^{\frac{\sqrt{3}}{3}}\frac{1}{\sqrt{1+t^2}}d\left(1+t^2\right)=\left[-\sqrt{1+t^2}\right]_{1}^{\frac{\sqrt{3}}{3}}=\left[-\sqrt{1+\frac{1}{3}}+\sqrt{2}\right]=\sqrt{2}-\frac{2}{\sqrt{3}}$;

(5) $\int_{-1}^{1}\frac{xdx}{\sqrt{5-4x}}\overset{\sqrt{5-4x}=t}{=\!=\!=}\int_{3}^{1}\frac{\frac{5-t^2}{4}\cdot\left(-\frac{1}{2}t\right)dt}{t}=-\frac{1}{8}\int_{3}^{1}\left(5-t^2\right)dt$

$$=-\frac{1}{8}\left[5t-\frac{1}{3}t^3\right]_3^1=-\frac{1}{8}\left[\frac{14}{3}-6\right]=\frac{1}{6}$$

(6) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos x\cos 2x dx=2\int_0^{\frac{\pi}{2}}\cos x\cos 2x\mathrm{d}x=\int_0^{\frac{\pi}{2}}\left(\cos 3x+\cos x\right)\mathrm{d}x$

$$=2\left[\frac{1}{3}\sin 3x+\sin x\right]_0^{\frac{\pi}{2}}=\frac{2}{3}.$$

(7) $\int_0^{\frac{\pi}{2}}\frac{dx}{3+2\cos x}\overset{u=\tan\frac{x}{2}}{=\!=\!=}\int_0^1\frac{1}{3+2\frac{1-u^2}{1+u^2}}\cdot\frac{2}{1+u^2}\mathrm{d}u=\int_0^1\frac{2}{5+u^2}\mathrm{d}u=\frac{2}{\sqrt{5}}\left[\arctan\frac{u}{\sqrt{5}}\right]_0^1$

$$=\frac{2}{\sqrt{5}}\left[\arctan\frac{1}{\sqrt{5}}-0\right]=\frac{2}{\sqrt{5}}\arctan\frac{1}{\sqrt{5}}.$$

(8) $\int_0^{\frac{\pi}{2}}\frac{\cos x}{6-5\sin x+\sin^2 x}dx=\int_0^{\frac{\pi}{2}}\frac{1}{6-5\sin x+\sin^2 x}d\sin x$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\left(\sin x-2\right)\left(\sin x-3\right)}d\sin x$$

$$=\int_0^{\frac{\pi}{2}}\left(\frac{1}{\sin x-3}-\frac{1}{\sin x-2}\right)d\sin x=\left[\ln\left|\frac{\sin x-3}{\sin x-2}\right|\right]_0^{\frac{\pi}{2}}=\ln\frac{4}{3};$$

(9) $\int_0^{\pi}(1-\sin^3 x)dx=\int_0^{\pi}dx-\int_0^{\pi}\sin^3 x dx=\pi+\int_0^{\pi}\left(1-\cos^2 x\right)d\cos x$

$$=\pi+\left[\cos x-\frac{1}{3}\cos^3 x\right]_0^{\pi}=\pi+\left[-1+\frac{1}{3}-1+\frac{1}{3}\right]=\pi-\frac{4}{3}.$$

(10) $\int_0^{\frac{\pi}{4}}\frac{1-\cos^4 x}{2}dx=\frac{1}{2}\int_0^{\frac{\pi}{4}}dx-\frac{1}{2}\int_0^{\frac{\pi}{4}}\cos^4 x dx=\frac{\pi}{8}-\frac{1}{2}\int_0^{\frac{\pi}{4}}\left(\frac{1+\cos 2x}{2}\right)^2 dx$

$$=\frac{\pi}{8}-\frac{1}{8}\int_0^{\frac{\pi}{4}}\left(1+2\cos 2x+\cos^2 2x\right)dx=\frac{\pi}{8}-\frac{1}{8}\int_0^{\frac{\pi}{4}}\left(1+2\cos 2x+\frac{1+\cos 4x}{2}\right)dx$$

$$=\frac{\pi}{8}-\frac{1}{16}\int_0^{\frac{\pi}{4}}\left(3+4\cos 2x+\cos 4x\right)dx=\frac{\pi}{8}-\frac{1}{16}\left[3x+2\sin 2x+\frac{1}{4}\sin 4x\right]_0^{\frac{\pi}{4}}$$

$$=\frac{\pi}{8}-\frac{1}{16}\left(\frac{3\pi}{4}+2\right)=\frac{5\pi}{64}-\frac{1}{8}.$$

(11) $\int_1^{e^2}\frac{dx}{x\sqrt{1+\ln x}}=\int_1^{e^2}\frac{d(1+\ln x)}{\sqrt{1+\ln x}}=\left[2\sqrt{1+\ln x}\right]_1^{e^2}=2\sqrt{3}-2;$

(12) $\int_1^{\ln 2}\sqrt{e^x-1}dx\overset{u=\sqrt{e^x-1}}{=\!=\!=}\int_{\sqrt{e-1}}^1 u\frac{2u}{1+u^2}du=2\int_{\sqrt{e-1}}^1\frac{(1+u^2)-1}{1+u^2}du$

$$=2\int_{\sqrt{e-1}}^1\left(1-\frac{1}{1+u^2}\right)du$$

$$=2\left[u-\arctan u\right]_{\sqrt{e-1}}^1=2\left(1-\frac{\pi}{4}-\sqrt{e-1}+\arctan\sqrt{e-1}\right)$$

$$=2-\frac{\pi}{2}-2\sqrt{e-1}+2\arctan\sqrt{e-1}.$$

2.计算下列定积分.

(1) $\int_0^1 xe^{-x}dx;$ (2) $\int_1^e x\ln x dx;$

(3) $\int_1^4\frac{\ln x}{\sqrt{x}}dx;$ (4) $\int_0^1 x\arctan x dx;$

(5) $\int_0^{\frac{\pi}{2}}e^x\cos x dx;$ (6) $\int_0^{\pi}(x\sin x)^2 dx;$

(7) $\int_1^e\sin(\ln x)dx;$ (8) $\int_0^{\frac{\pi}{2}}\cos^8 x dx.$

解(1) $\int_0^1 xe^{-x}dx=-\int_0^1 xde^{-x}=\left[-xe^{-x}\right]_0^1-\int_0^1 e^{-x}d(-x)=-e^{-1}-\left[e^{-x}\right]_0^1=-2e^{-1}+1;$

(2) $\int_1^e x\ln x\mathrm{d}x=\int_1^e\ln x\mathrm{d}\left(\frac{1}{2}x^2\right)=\frac{1}{2}\left[x^2\ln x\right]_1^e-\frac{1}{2}\int_1^e x^2\mathrm{d}(\ln x)$

$$=\frac{1}{2}e^2-\frac{1}{2}\int_1^e x\mathrm{d}x=\frac{1}{2}e^2-\frac{1}{4}\left[x^2\right]_1^e=\frac{1}{4}e^2+\frac{1}{4}.$$

(3) $\int_1^4\frac{\ln x}{\sqrt{x}}\mathrm{d}x=\int_1^4\ln x\mathrm{d}\left(2\sqrt{x}\right)=\left[2\sqrt{x}\ln x\right]_1^4-\int_1^4 2\sqrt{x}\cdot\frac{1}{x}\mathrm{d}x$

$$=2\sqrt{4}\ln 4-0-4\left[\sqrt{x}\right]_1^4=8\ln 2-4.$$

(4) $\int_0^1 x\arctan x\mathrm{d}x=\frac{1}{2}\int_0^1\arctan x\mathrm{d}\left(x^2\right)=\frac{1}{2}\left[x^2\arctan x\right]_0^1-\frac{1}{2}\int_0^1\frac{x^2}{1+x^2}\mathrm{d}x$

$$=\frac{\pi}{8}-\frac{1}{2}\left[x-\arctan x\right]_0^1=\frac{\pi}{4}-\frac{1}{2}.$$

(5) $\int_0^{\frac{\pi}{2}}e^x\cos x\mathrm{d}x=\int_0^{\frac{\pi}{2}}e^x\mathrm{d}(\sin x)=\left[e^x\sin x\right]_0^{\frac{\pi}{2}}-\int_0^{\frac{\pi}{2}}\sin x\mathrm{d}\left(e^x\right)$

$$= \mathrm{e}^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \mathrm{e}^x \sin x \mathrm{d}x = \mathrm{e}^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \mathrm{e}^x \mathrm{d}\left(\cos x\right) = \mathrm{e}^{\frac{\pi}{2}} + \left[\mathrm{e}^x \cos x\right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \mathrm{e}^x \cos x \mathrm{d}x$$

$$= \mathrm{e}^{\frac{\pi}{2}} - 1 - \int_0^{\frac{\pi}{2}} \mathrm{e}^x \sin x \mathrm{d}x$$

所以 $\int_0^{\frac{\pi}{2}} \mathrm{e}^x \cos x \mathrm{d}x = \frac{1}{2}\left(\mathrm{e}^{\frac{\pi}{2}} - 1\right).$

(6) $\int_0^{\pi} (x \sin x)^2 \mathrm{d}x = \frac{1}{2}\int_0^{\pi} x^2 \left(1 - \cos 2x\right) \mathrm{d}x = \frac{\pi^3}{6} - \frac{1}{4}\int_0^{\pi} x^2 \mathrm{d}\left(\sin 2x\right)$

$$= \frac{\pi^3}{6} - \frac{1}{4}\left[x^2 \sin 2x\right]_0^{\pi} + \frac{1}{2}\int_0^{\pi} x \sin 2x \mathrm{d}x = \frac{\pi^3}{6} - \frac{1}{4}\int_0^{\pi} x \mathrm{d}\left(\cos 2x\right)$$

$$= \frac{\pi^3}{6} - \frac{1}{4}\left[x \cos 2x\right]_0^{\pi} + \frac{1}{4}\int_0^{\pi} \cos 2x \mathrm{d}x = \frac{\pi^3}{6} - \frac{\pi}{4}.$$

(7) $\int_1^{\mathrm{e}} \sin(\ln x) \mathrm{d}x \overset{\ln x = t}{=\!=\!=} \int_0^1 \mathrm{e}^t \sin t \mathrm{d}t = \int_0^1 \sin t \mathrm{d}\left(\mathrm{e}^t\right) = \left[\mathrm{e}^t \sin t\right]_0^1 - \int_0^1 \mathrm{e}^t \cos t \mathrm{d}t$

$$= \mathrm{e} \sin 1 - \int_0^1 \cos t \mathrm{d}\left(\mathrm{e}^t\right) = \mathrm{e} \sin 1 - \left[\mathrm{e}^t \cos t\right]_0^1 - \int_0^1 \mathrm{e}^t \sin t \mathrm{d}t$$

$$= \mathrm{e} \sin 1 - \mathrm{e} \cos 1 + 1 - \int_0^1 \mathrm{e}^t \sin t \mathrm{d}t,$$

所以 $\int_1^{\mathrm{e}} \sin(\ln x) \mathrm{d}x = \frac{\mathrm{e}}{2}\left(\sin 1 - \cos 1\right) + \frac{1}{2}.$

(8) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^8 x \mathrm{d}x = 2\int_0^{\frac{\pi}{2}} \cos^8 x \mathrm{d}x = 2 \cdot \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{35}{128}\pi.$

3.利用函数的奇偶性计算下列积分.

(1) $\int_{-\pi}^{\pi} x^6 \sin x dx;$ (2) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4\cos^4 \theta d\theta;$

(3) $\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}} dx;$ (4) $\int_{-5}^{5} \frac{x^4 \sin x}{x^4 + 2x^2 + 1} dx.$

解 (1) 因为 $f(x) = x^6 \sin x$ 为奇函数，因此 $\int_{-\pi}^{\pi} x^6 \sin x \mathrm{d}x = 0.$

(2) 因为 $f(x) = 4\cos 4\theta$ 为偶函数，因此

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4\cos^4 \theta d\theta = 2\int_0^{\frac{\pi}{2}} 4\cos^4 \theta d\theta = 8\int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta = 8 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{2};$$

(3) 因为 $f(x) = \frac{(\arcsin x)^2}{\sqrt{1-x^2}}$ 为偶函数，因此

$$\int_{-\frac{1}{2}}^{\frac{1}{2}}\frac{(\arcsin x)^2}{\sqrt{1-x^2}}\mathrm{d}x=2\int_0^{\frac{1}{2}}(\arcsin x)^2\,\mathrm{d}(\arcsin x)=\frac{2}{3}\Big[(\arcsin x)^3\Big]_0^{\frac{1}{2}}=\frac{\pi^2}{324}.$$

(4) 因为 $f(x)=\dfrac{x^4\sin x}{x^4+2x^2+1}$ 为奇函数，因此 $\int_{-5}^{5}\dfrac{x^4\sin x}{x^4+2x^2+1}\mathrm{d}x=0.$

4.证明下列等式.

（1） $\int_0^1 x^m(1-x)^n\,dx=\int_0^1 x^n(1-x)^m\,dx\quad(m>0,n>0);$

（2） $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx;$

(3) $\int_0^a f(x^2)dx=\dfrac{1}{2}\int_{-a}^{a}f(x^2)dx.$

证（1） $\int_0^1 x^m(1-x)^n\,\mathrm{d}x\overset{1-x=t}{=\!=\!=}-\int_1^0(1-t)^n t^m\,\mathrm{d}t=\int_0^1 t^m(1-t)^n\,\mathrm{d}t=\int_0^1 x^m(1-x)^n\,\mathrm{d}x.$

（2） $\int_a^b f(x)\mathrm{d}x\overset{x=a+b-t}{=\!=\!=}-\int_b^a f(a+b-t)\mathrm{d}t=\int_a^b f(a+b-t)\mathrm{d}t=\int_a^b f(a+b-x)\mathrm{d}x.$

（3）因为

$$\int_{-a}^{a}f(x^2)\,\mathrm{d}x=\int_{-a}^{0}f(x^2)\,\mathrm{d}x+\int_0^a f(x^2)\,\mathrm{d}x.$$

而右端第一个积分为

$$\int_{-a}^{0}f(x^2)\,\mathrm{d}x\overset{x=-t}{=\!=\!=}-\int_a^0 f(t^2)\,\mathrm{d}t=\int_0^a f(t^2)\,\mathrm{d}t=\int_0^a f(x^2)\,\mathrm{d}x,$$

故

$$\int_{-a}^{a}f(x^2)\mathrm{d}x=\int_{-a}^{0}f(x^2)\mathrm{d}x+\int_0^a f(x^2)\mathrm{d}x=2\int_0^a f(x^2)\mathrm{d}x,$$

即

$$\int_0^a f(x^2)\mathrm{d}x=\frac{1}{2}\int_{-a}^{a}f(x^2)\mathrm{d}x.$$

5.若函数 $y=f(x)$ 在 $[0,1]$ 上连续，证明

（1） $\int_0^{\frac{\pi}{2}}f(\sin x)dx=\int_0^{\frac{\pi}{2}}f(\cos x)dx;$

（2） $\int_0^{\pi}xf(\sin x)dx=\dfrac{\pi}{2}\int_0^{\pi}f(\sin x)dx.$

证（1） $\int_0^{\frac{\pi}{2}}f(\sin x)dx\overset{x=\frac{\pi}{2}-t}{=\!=\!=}-\int_{\frac{\pi}{2}}^0 f(\cos\mathrm{t})dt=\int_0^{\frac{\pi}{2}}f(\cos x)dx.$

（2） $\int_0^{\pi}xf(\sin x)dx\overset{x=\pi-t}{=\!=\!=}\int_{\pi}^0(\pi-t)f(\sin(\pi-t))d(\pi-t)=\int_0^{\pi}(\pi-t)f(\sin t)dt$

$=\int_0^{\pi}\pi f(\sin t)dt-\int_0^{\pi}tf(\sin t)dt=\pi\int_0^{\pi}f(\sin x)dx-\int_0^{\pi}xf(\sin x)dx$

故 $\int_0^{\pi} xf(\sin x)dx = \frac{\pi}{2}\int_0^{\pi} f(\sin x)dx.$

6.设函数 $f(x)$ 在区间 $[0,1]$ 上连续，证明

$$\int_0^1[\int_0^x f(t)dt]dx = \int_0^1(1-x)f(x)dx.$$

证 由分部积分公式，有

$$\int_0^1\left[\int_0^x f(t)\,\mathrm{d}t\right]\mathrm{d}x=\left[x\int_0^x f(t)\,\mathrm{d}t\right]_0^1-\int_0^1 xf(x)\mathrm{d}x=\int_0^1 f(t)\,\mathrm{d}t-\int_0^1 xf(x)\,\mathrm{d}x$$

$$=\int_0^1 f(x)\,\mathrm{d}x-\int_0^1 xf(x)\,\mathrm{d}x=\int_0^1(1-x)f(x)\,\mathrm{d}x\,.$$